Product Requirements Document

[Working name: CODA] — Marketplace-First Commerce & Embedded Finance Platform for Nigerian SMEs

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1. Vision

A Nigerian micro/small business owner should be able to start selling online in under 5 minutes with nothing but a phone number — no smartphone required, no data plan required, no paperwork required on day one. As they use the platform to sell, invoice, and record transactions, they build a visible, portable credit history that unlocks real bank-backed working capital — without ever filling out a loan application.

The one-line pitch: *Selar/Bumpa's storefront simplicity, reaching the traders those apps don't — starting on USSD and WhatsApp, not just an app — with a credit score that grows automatically the more you sell.*

2. Problem Statement

- Nigeria has ~40M MSMEs; 80%+ operate informally and lack access to formal credit because traditional lending requires collateral, audited accounts, and documentation most don't have.
- Existing storefront tools (Bumpa, Selar) and bookkeeping tools (Kippa) solve pieces of this, but assume a smartphone, consistent data access, and app literacy — which excludes a large tier of traders who sell primarily through word-of-mouth, market stalls, and WhatsApp/Instagram DMs.
- Bumpa has now launched embedded lending (Bumpa Capital, via Vendorcredit) using on-platform sales data — validating the model, but only for the smartphone-first segment they already serve.
- Kippa's 2024 collapse (users locked out of their own transaction/inventory data after a banking-license pivot) shows the failure mode to avoid: owning banking risk directly, and treating the ledger as disposable.

3. Target Users (in priority order)

1. **The unregistered micro-trader** — provisions store, fashion/tailoring, food vendor, phone accessories. Sells via WhatsApp/Instagram + physical location. Owns a basic-

to-mid Android phone, unreliable data. Has never used a bookkeeping app. This is the primary wedge user.

2. **The semi-formal small business —** has a CAC-registered name, maybe a Bumpa/Paystack storefront already, wants better records and access to capital they currently can't get from a bank.
3. **The bank/MfB partner** (a second customer, not an end user) — needs a clean, auditable, CBN-compliant data feed and risk signal to originate small loans profitably.

4. Product Principles

- **No feature ships that requires a smartphone or constant data to be usable at a basic level.** USSD/WhatsApp parity for the core loop (sell → record → get paid → build score) is a hard requirement, not a stretch goal.
- **The ledger is sacred.** Whatever else changes, a merchant's sales/inventory/debtor records must never become inaccessible. (Direct response to Kippa's failure.)
- **We never hold customer deposits or originate loans ourselves.** We are the UX + data layer; a licensed bank/MfB partner carries balance-sheet risk. This determines the entire technical and legal architecture.
- **Every screen is usable by someone who has never used a business app before.** No jargon — "money in / money out," not "revenue / COGS."
- **The reward loop must be visible within days, not months —** score movement, unlocked features, and loan eligibility should feel like leveling up, not paperwork.

5. Competitive Positioning Summary

	Bumpa	**Selar**	**Kippa (defunct)**	**Us**
Storefront/marketplace	Yes	Digital products only	Basic	Yes
Bookkeeping	Yes	No	Yes (lost)	Yes
Embedded lending	Yes (via Vendorcredit)	No	Attempted, collapsed	Yes (via BaaS partner)
USSD access	No	No	No	**Yes — our wedge**

	Bumpa	Selar	Kippa (defunct)	Us
WhatsApp-native commerce	Partial (Meta integration in progress)	No	No	**Yes, deep**
Offline-first records	No	No	No	**Yes**

We are not trying to out-feature Bumpa on the smartphone-first segment. We win the segment below it, then move upmarket as those merchants grow.

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6. Roadmap Overview

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Phase 0   Marketplace MVP (web + WhatsApp)              Months 1-2
Phase 1   Ledger: receipts, invoices, inventory          Months 2-3
Phase 2   Identity & tiered KYC                          Months 3-4
Phase 3   Credit score v1 (rules-based)                  Months 4-5
Phase 4   BaaS partnership + embedded lending pilot       Months 5-7
Phase 5   USSD channel + offline-first PWA                Months 6-8
(parallel with 4)
Phase 6   Open Banking data pull-in + score v2 (model)     Months 8-10
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Timeline assumes solo/near-solo development around a full-time job — treat months as "focused iterations," not calendar guarantees. Sequencing matters more than dates: **nothing in Phase 2+ should block Phase 0 shipping and getting real merchants using it.**

PHASE 0 — Marketplace MVP

Goal: A business owner can register, get a shareable storefront link, list products/services with pricing, and receive orders — in under 5 minutes, on any device.

7.1 User Stories

- As a new user, I can sign up with just my phone number (OTP) and business name — no BVN/NIN required yet.
- As a merchant, I can create a storefront with a custom slug (`coda.ng/store/mama-ngozi`) in under 3 steps.
- As a merchant, I can add a product/service with a photo (or skip photo), price, and optional stock count.

- As a merchant, I can share my storefront link directly to WhatsApp status, or share my catalog as a native WhatsApp Business catalog.
- As a customer, I can browse a storefront on a slow connection without it timing out or costing excessive data.
- As a merchant, I can see a simple "orders" inbox — new order, paid, fulfilled — with no more than 3 states.
- As a merchant, I can accept payment via Paystack (card, bank transfer, USSD-based bank payment) without integrating anything myself.

7.2 Functional Requirements

ID	Requirement	Priority
F0.1	Phone-number + OTP registration (no email required)	P0
F0.2	Business profile: name, category, location (state/LGA), optional logo	P0
F0.3	Storefront generator: unique slug, mobile-first, <200KB initial page weight	P0
F0.4	Product/service CRUD: name, price, photo (optional, compressed client-side before upload), stock count (optional)	P0
F0.5	Order flow: customer selects item(s) → checkout via Paystack → order appears in merchant inbox	P0
F0.6	WhatsApp catalog sync (share-to-WhatsApp deep link at minimum; native Catalog API integration as stretch)	P0
F0.7	Order status states: New → Paid → Fulfilled/Cancelled	P0
F0.8	Basic storefront themes (2-3 templates, not customizable CSS — keep it simple)	P1
F0.9	Multi-language toggle (English + Pidgin copy at minimum)	P2

7.3 Non-Functional Requirements

- Storefront pages must render usable content in under 3 seconds on a 3G-equivalent connection (target: Lighthouse mobile score >85, no render-blocking JS for the storefront view itself).
- No feature in Phase 0 may require the merchant to have used a computer before — mobile web is the primary surface, not desktop.
- All product images auto-compressed/resized on upload (client-side, before hitting Supabase storage) to keep both merchant upload cost and customer browsing cost

low.

7.4 Out of Scope for Phase 0

Banking, KYC beyond phone OTP, credit scoring, USSD, inventory alerts, staff accounts, multi-currency. Resist scope creep here — this phase's only job is proving people will register and share their storefront link.

7.5 Success Metrics

- 100 registered storefronts with at least 1 product listed within first 6 weeks of launch.
- ⩾30% of registered merchants share their storefront link at least once (tracked via link clicks).
- ⩾15% of storefronts receive at least one real order within 30 days of creation.

PHASE 1 — Ledger: Receipts, Invoices, Inventory

Goal: Every transaction a merchant makes — on-platform or off — gets recorded in one place, becoming the foundation of their eventual credit file.

8.1 User Stories

- As a merchant, I can generate a professional receipt/invoice (PDF or shareable image) for any sale, including ones that didn't happen through my storefront (manual entry).
- As a merchant, I can log an expense in two taps (amount + category).
- As a merchant, I can see a simple weekly "money in / money out" summary — no accounting terms.
- As a merchant, I can track who owes me money (informal customer credit/debt) and mark it paid.
- As a merchant, my records are exportable at any time (PDF/CSV) — this is a trust commitment, stated explicitly in-product.

8.2 Functional Requirements

ID	Requirement	Priority
F1.1	Manual sale entry (amount, item optional, customer optional)	P0
F1.2	Auto-logging of on-platform storefront orders into the same ledger	P0
F1.3	Receipt/invoice generator — branded, downloadable, shareable via WhatsApp	P0

ID	Requirement	Priority
F1.4	Expense logging (amount, category dropdown: stock, transport, rent, utilities, other)	P0
F1.5	Debtor tracking ("who owes me") with due date and payment reminder trigger	P0
F1.6	Simple stock/inventory count synced with storefront listings	P1
F1.7	Weekly digest (via WhatsApp message or in-app) — money in, money out, top-selling item	P1
F1.8	One-tap full data export (CSV + PDF) — always available, never gated	P0
F1.9	Receipt-from-photo: merchant photographs a paper receipt/handwritten note, OCR extracts amount/date as a draft entry for confirmation	P2

8.3 Non-Functional Requirements

- All ledger writes must succeed locally first and sync when connectivity returns (offline-first) — see Phase 5 for full architecture, but the data model must be designed for this from Phase 1, not retrofitted later.
- Export must never be paywalled or delayed — this is the anti-Kippa commitment and should be stated in the product's terms.

8.4 Success Metrics

- ⩾40% of active merchants log at least 3 transactions/week.
- ⩾20% of merchants generate at least one invoice/receipt through the platform monthly.
- Data export feature used by <5% of users (a proxy for trust — high usage would suggest people are extracting data because they don't trust the platform to keep it).

PHASE 2 — Identity & Tiered KYC

Goal: Progressive verification that unlocks capability, without gating basic usage — compliant with the CBN's 2026 tiered KYC and Baseline Standards for automated AML.

9.1 Tier Structure

Tier	Trigger	Requirements	Unlocks	Daily txn limit
Tier 0	Signup	Phone number only	Storefront, product listing, ledger (manual)	N/A — no money movement
Tier 1	First real payment collection	BVN **or** NIN, verified in real time via NIBSS/NIMC (not self-declared)	Accept customer payments on-platform, receive weekly digest	₦30,000
Tier 2	Merchant opts in for credit features	BVN **and** NIN verified, government ID + address, liveness check	Score visibility, first micro-loan tier, higher transaction limit	₦500,000
Tier 3	Business growth / larger loan request	CAC registration, beneficial ownership (⩾5%) disclosure, business bank statement (via Open Banking consent)	Full loan thresholds, business overdraft	Unlimited

9.2 Functional Requirements

ID	Requirement	Priority
F2.1	Real-time BVN/NIN verification via a licensed identity API vendor (Dojah, Youverify, or Smile ID — evaluate on cost-per-verification and NIBSS/NIMC integration maturity)	P0
F2.2	Liveness/selfie check for Tier 2+	P0
F2.3	In-app/in-chat prompts nudging Tier 0→1 users toward verification, tied to a visible benefit ("Verify to unlock payments") — never blocking core usage	P0
F2.4	Consent management system: explicit, time-bound, revocable consent logging for every data-sharing action (NDPA 2023 requirement)	P0
F2.5	CAC lookup/verification integration for Tier 3	P1

ID	Requirement	Priority
F2.6	Audit trail: every KYC decision and data access logged and retrievable for CBN/NDPC audit requests	P0

9.3 Compliance Notes

- Self-declaration alone does not satisfy any tier under the March 2026 Baseline Standards — this is a hard build requirement, not a policy nice-to-have.
- Risk profiling must update through the customer lifecycle, not just at onboarding — plan for a periodic re-screening job, not a one-time check.
- This phase requires either an in-house compliance advisor or a fractional consultant before launch — do not self-interpret CBN circulars for production KYC flows.

9.4 Success Metrics

- ⩾60% of merchants who reach Tier 0 convert to Tier 1 within 30 days.
- Verification API false-rejection rate <5% (tracked via support complaints/appeals).

PHASE 3 — Credit Score v1 (Rules-Based)

Goal: A transparent, explainable score that starts at 0 and moves visibly with good behavior — built from rules, not a trained model, because you won't have outcome data yet.

10.1 Score Inputs (v1, rules-based, weighted)

- Sales consistency (transactions logged per week, variance over time)
- Revenue trend (rolling 4-week growth/decline)
- Invoice-to-payment turnaround time
- Debtor repayment rate (of tracked customer debts, % paid on time)
- Platform tenure (account age, weighted lightly)
- KYC tier (higher tier = eligible for higher score ceiling, not a bonus itself)
- (Phase 6+) Off-platform bank transaction history via Open Banking consent

10.2 Functional Requirements

ID	Requirement	Priority
F3.1	Score displayed prominently in-app and in WhatsApp digest, 0-100 scale, in plain language ("Building," "Good," "Strong") not raw jargon	P0

ID	Requirement	Priority
F3.2	Score recalculated weekly, with a visible "what changed" explanation each time	P0
F3.3	Explainability log: for any score, a merchant (or auditor/partner bank) can see the factors that produced it	P0
F3.4	Score gating tied explicitly to loan tiers (see Phase 4)	P0

10.3 Notes

- Do not market this as "AI" until there's a trained model on real repayment outcomes (Phase 6+). CBN's automated AML/risk standards expect auditable, explainable decisions — a rules engine is actually the *safer* and more defensible choice at this stage, not a lesser one.

10.4 Success Metrics

- ⩾50% of Tier 1+ merchants check their score at least twice a month.
- Score-to-loan-approval correlation tracked from first pilot cohort (Phase 4) to validate/recalibrate weights.

PHASE 4 — BaaS Partnership & Embedded Lending Pilot

Goal: A merchant with a qualifying score can request and receive a small loan without leaving the platform, disbursed and serviced by a licensed partner.

11.1 Functional Requirements

ID	Requirement	Priority
F4.1	Partner integration (BaaS provider or direct MfB) for loan origination, disbursement, and repayment collection	P0
F4.2	In-app loan offer screen: amount, tenor, repayment schedule, total cost — shown in plain terms (naira amounts, not APR jargon)	P0
F4.3	Auto-repayment via deduction from platform sales (mirrors Bumpa Capital's model) with manual repayment fallback for off-platform sales	P0
F4.4	Loan status tracker (visible progress bar-style UI)	P0
F4.5	Post-repayment score boost + threshold increase, communicated immediately	P0

ID	Requirement	Priority
F4.6	Default/late-payment handling flow — compliant, non-predatory, clearly disclosed before borrowing	P0

11.2 Business/Legal Requirements

- Formal BaaS or MfB partnership agreement — this is a business-development track that should start in parallel with Phase 2, not after Phase 3 is built, because partner onboarding/compliance review takes months.
- Loan terms, interest, and default handling must be reviewed by legal counsel familiar with CBN consumer lending disclosure rules before pilot launch.

11.3 Success Metrics

- Pilot cohort: 50-100 Tier 2+ merchants, first loan sizes ₦5,000-₦20,000.
- Repayment rate ⩾90% within agreed tenor for pilot cohort (this is the number that determines whether the partner bank scales with you).

PHASE 5 — USSD Channel & Offline-First PWA

Goal: The core loop (record a sale, check balance/score, confirm payment, request a small loan) works on any phone, with or without data.

12.1 USSD Scope (deliberately narrow — max 3 menu levels)

1. Check sales summary (today/week)
2. Record a sale (amount only, optional item code)
3. Check credit score / loan eligibility
4. Request a receipt via SMS
5. Check/request a loan (Tier 2+ only)

12.2 Functional Requirements

ID	Requirement	Priority
F5.1	USSD shortcode leased via aggregator (Africa's Talking, Interswitch, or Flutterwave USSD)	P0
F5.2	Session design capped at 3 menu levels, tested for abandonment rate	P0

ID	Requirement	Priority
F5.3	SMS fallback for receipts/confirmations where USSD session can't display full content	P0
F5.4	WhatsApp Business API integration: catalog, structured button/list replies (not free text) for record-sale, check-score, get-invoice flows	P0
F5.5	PWA offline mode: service worker caching for read views (past sales, inventory), local-first writes with background sync queue	P0
F5.6	Conflict resolution strategy for offline writes that sync late (e.g., stock count desync)	P1

12.3 Success Metrics

- USSD session completion rate ⩾70% (proxy for menu design being genuinely usable).
- ⩾25% of merchants in low-connectivity areas (self-reported or inferred from sync-lag data) use USSD or offline-PWA mode at least weekly.

PHASE 6 — Open Banking Data Pull-In & Score v2

Goal: Pull a merchant's existing bank transaction history (with consent) to build a credit picture from day one instead of waiting months for on-platform data, and move from rules-based to a trained scoring model once real repayment outcomes exist.

13.1 Functional Requirements

ID	Requirement	Priority
F6.1	Register as API Consumer on CBN's Open Banking Registry	P0
F6.2	Consent flow: explicit, time-bound, revocable, per NDPA 2023 — this becomes a formal legal/UX artifact, not just a checkbox	P0
F6.3	Ingest consented bank transaction history into score model as an input alongside on-platform data	P1
F6.4	Trained model v2 using Phase 4 pilot repayment outcomes, replacing/augmenting rules engine — with explainability preserved	P1

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14. Cross-Cutting Non-Functional Requirements

- **Data residency & security**: all customer financial data handled per NDPA 2023; encryption at rest and in transit; access logging for every read of sensitive fields (BVN, NIN, transaction history).
- **Ledger durability**: automated backups, tested restore process, explicit uptime/data-availability commitment communicated to users — this is a direct product commitment in response to the Kippa failure, and should appear in your terms of service, not just your infrastructure.
- **Language/literacy**: no financial jargon anywhere in the primary UI copy; every technical term has a plain-language equivalent.
- **Cost-to-serve**: track infra cost per active merchant (image storage, USSD session fees, identity verification API calls) from Phase 0 — this determines pricing model viability long before you reach lending revenue.

15. Risks & Mitigations

Risk	Mitigation
Bumpa/Kippa-style incumbents move down-market into USSD/low-data before you scale	Ship Phase 0-1 fast, focus distribution in underserved geographies (outside Lagos/Abuja core), let the USSD wedge be the moat, not a delayed feature
BaaS/MfB partner negotiation takes longer than expected	Start partner conversations in Phase 2, not Phase 4; have 2-3 candidate partners in parallel
Identity verification costs eat margin at scale	Negotiate volume pricing early; consider Tier 0 remaining genuinely free/unlimited to maximize top-of-funnel before verification costs kick in
Regulatory interpretation risk (CBN circulars are dense and evolving)	Budget for a compliance consultant before Phase 2 ships, not after an issue arises
Ledger trust failure (a Kippa repeat)	Explicit uptime/export commitments in ToS; treat data durability as a P0 engineering requirement, never deprioritized for feature velocity

16. Open Questions (resolve before Phase 2)

- Which identity verification vendor (Dojah vs Youverify vs Smile ID) — needs a cost/coverage comparison specific to your expected volume.

- Which BaaS/MfB partner is realistically reachable for a solo founder — this may determine your actual Phase 4 timeline more than anything technical.
- Pricing model: free storefront + transaction fee (like Paystack Storefront) vs. subscription (like Bumpa) vs. take-rate on loan facilitation — needs its own analysis before Phase 0 monetization decisions lock in.